

Controlling Visual and Auditory Information Propagation Using Interactive AR Partitions in Open-Plan Office

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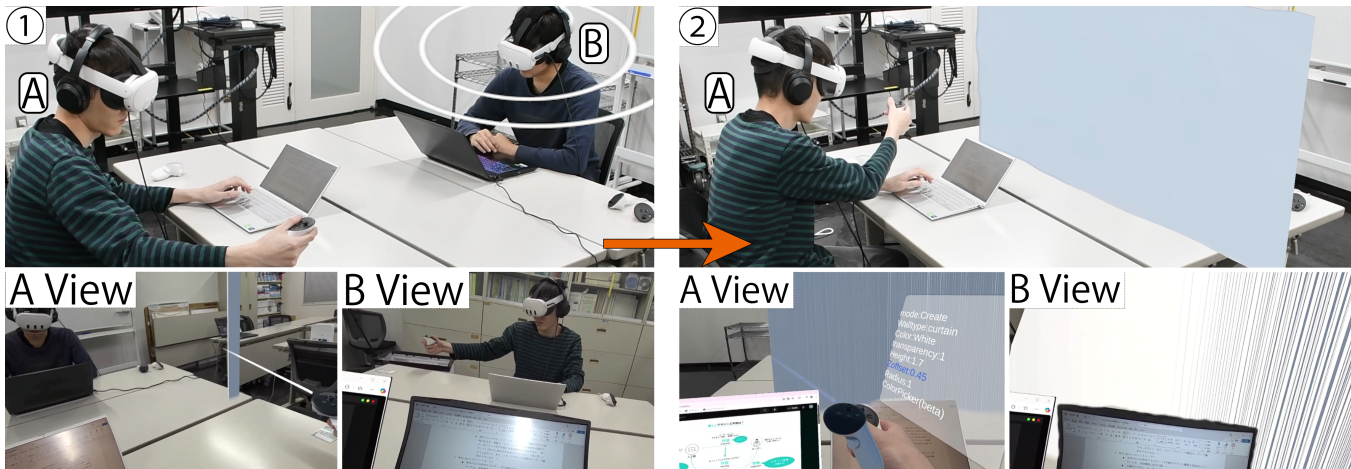


Figure 1: A use case of the AR partition system: (1) User A blocks visual and auditory noise emitted by nearby User B by creating an AR partition. (2) The partition separates User A from User B and reduces User B's voice volume level for User A.

Abstract

Recently, open-plan offices have often been used in the company and workspaces, but those offices are insufficient in security and privacy because much of the information is unintentionally available to unrelated persons. In this work, we propose an interactive AR partitioning system that allows users to visually and auditory divide or connect working spaces on an AR headset and a noise-canceling headphone setup. We design and implement a first proof-concept prototype using MR HMD's video passthrough function and sound/noise-controllable headphones. This interactive system showcases various partitioning scenarios, highlighting its flexibility and effectiveness in managing both individual and group activities in the future of open-plan offices.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality; Auditory feedback.**

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Keywords

Noise-canceling headphones, privacy, space design

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1 Introduction

Recently, open-plan offices have become widely adopted in companies' workspaces. One of the advantages of such offices is facilitating active communication among employees. However, information displayed on monitors can be easily viewed by unintended persons, and conversations are often overheard by unrelated individuals nearby. Traditional open-plan workspaces are not designed to manage this unintentional visual and auditory information propagation, which negatively impacts workspace security, employee privacy, and overall satisfaction with the working environment.

Nowadays, AR (Augmented Reality) technology is rapidly advancing, and high-resolution AR glasses or mixed reality HMDs with high-quality color passthrough are commercially available. Also, people increasingly use noise-canceling headphones on a daily basis. According to such circumstances, we assume that HMD and noise-canceling headphones would be ordinary tools in the office.

Thus, we explore ways to control visual and auditory distractions by using these tools.

In this paper, we propose an interactive AR partitioning system that enables users to visually and auditorily divide or connect workspaces by creating user-defined AR partitions. The AR partitions can be generated and displayed through AR devices; in our implementation, we use color pass-through VR HMDs and noise-canceling headphones. While working on virtual content within an HMD or viewing actual computer screens, users can quickly create AR partitions in their field of view (Figure 1) to block visual distractions in front of them and suppress surrounding auditory noise (Figure 1, A View). Additionally, ambient sound capture is automatically adjusted according to the size of the generated partitions, further controlling surrounding noise. This system enables users to instantly create private, quiet virtual rooms anywhere in the office. Moreover, if multiple users have the same setup, they can share their generated AR partitions and experience identical information propagation features within a shared AR environment (Figure 1, B View).

We implemented a prototype and demonstrated its novel and effective interactive capabilities through representative application scenarios that highlight the flexibility and uniqueness of the AR partitions.

2 Related Work

2.1 Traditional and Virtual Workspaces

The advantages and disadvantages of open-plan offices have been the subject of ongoing debate, and numerous researchers have worked on these issues over the years. For instance, Mary compared traditional office spaces with open-plan offices, highlighting their respective merits and drawbacks[14]. Recently, Lee et al. investigated how the arrangement of partitions in open-plan offices affects speech privacy. They compared different partition layouts and explored methods to enhance speech privacy while maintaining effective communication[7].

As Fujita et al. proposed, the physical office spaces and virtual workspaces are being merged[2], and virtual reality technology has been introduced in the office design. Lee et al. explored the use of AR peripheral aerial displays[5] and found that AR-based environment switching improves the efficiency and flexibility of shared space.

Recently, metaverse office applications have been developed, enabling multiple users to conduct meetings and other activities remotely. Ruvimova et al. [10] investigated the performance of work efficiency in VR-based workspaces compared to non-VR workspaces. They reported that the performance in open VR spaces was superior to that in non-VR open-plan offices.

2.2 Virtually and Physically Augmented Office

To make the office workspace more functional, several augmented workspace concepts have been proposed. Takeuchi tackled the auditory noise problem and proposed a system that propagates people's conversation in the space to only people staying within a predefined area[11]. However, this method is designed for controlling auditory noise, not for managing visual noise.

Lee et al. addressed office visual noise with on-desk AR partitions displayed through optical see-through AR glasses [6]. Their method reduced visual distractions and improved user performance by allowing AR partitions to be visible to others. However, it was not aimed to address auditory noise.

Onishi et al. addressed visual noise in workspaces with interactive robotic partitions [8]. Similarly, Itani et al. proposed self-dispersing acoustic robot swarms [4], and Fukushima et al. employed automatic blind-type partitions [3]. While these methods allow dynamic physical partitions, they involve high preparation costs and limited adaptability due to fixed physical features.

Wu and Yang demonstrated the use of virtual shields[12] or guardians[13] to address the coexistence of VR and non-VR users. While this system employs virtual partitions shared among multiple VR users, its primary focus is ensuring the safe use of VR content rather than dynamically managing office spaces.

3 AR Partition System Prototype

Based on the existing practices in interior domains, we implemented a prototype of the AR partition system that allows users to edit various partition forms. This prototype is based on the assumption that AR glasses will become a standard tool for creating virtual workspaces for both individual and group activities. To simulate this scenario affordably, we utilized a color passthrough VR HMD, which offers clear situational awareness of the real world along with a wide field of view for VR content.

3.1 Setup

A schematic diagram of our AR partition system is shown in Figure 2. We used a Meta Quest3 HMD and Sony-WH-1000XM3 headphones (ASC headphone) for each user. We used Unity(2023.2.18f1) to implement AR render software. The user sees the surroundings via the color pass-through function of the Meta Quest 3 (2,064 x 2,208 dots per eye with up to 120 Hz refresh rates). When no partition is in the space, the headphone's ambient sound level is set to max, allowing the user to hear the surround sound clearly through the headphones. If the partitions are generated on the VR renderer, the ambient sound level is adjusted. To implement a multi-user experience, we utilized Normcore; this plugin employs the RealtimeAPI, a WebRTC-based communication protocol, offering low-latency, fully encrypted communication for enhanced security.

3.2 Workflow

The Meta Quest 3 headset and our VR renderer are connected via Quest Link. When a partition create command is sent, the VR renderer renders a requested AR partition at the specified statuses. This AR partition is then displayed in the AR space on the Meta Quest 3 (Figure 1 (2)). The AR renderer interacts with the Headphone API to adjust the ambient sound capture (ASC) level. The ASC level is then applied to the pass-through headphones.

The communication among multiple AR partition systems is achieved through the Normcore Cloud Server. By the position, rotation, and size of users and partitions are exchanged via the Normcore Cloud Server, allowing multiple users to join the AR workspaces and utilize the AR partitions collaboratively.

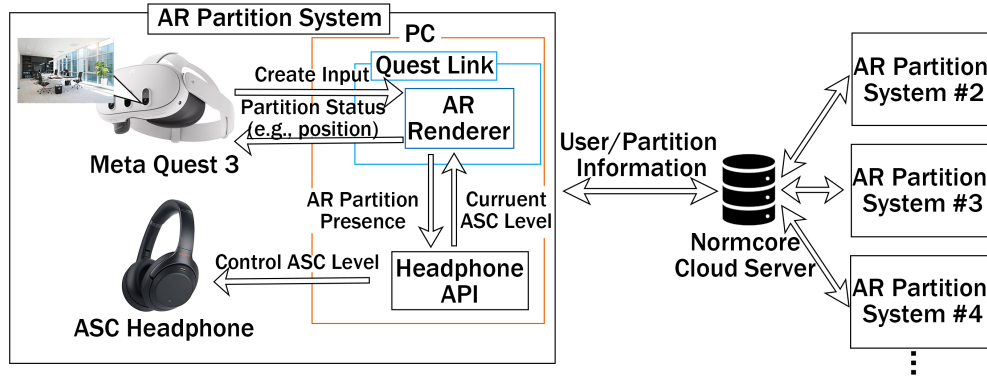


Figure 2: System overview of AR partition System

3.3 Audio control

We utilized the Sony-Headphone Client[9]. By operating this API through PyAutoGUI, the system dynamically adjusts the ambient sound capture level. The headphones can reduce noise by about 5 ~ 10db through the noise cancellation mode, according to our pilot study.

3.4 User Interface

The user interface menu is displayed above the rendered right-hand controller in the AR space (Figure 3 (a)). It displays items such as mode, wall type, color, transparency, wall height, depth offset from the controller, circular wall radius, and color picker. Each item is selected by up/down operation of the joystick, and its status can be switched by left/right stick operation. The current status can be confirmed on the UI in real time. To create a partition, the user first checks if the create bar is shown, then defines the height and off-set of the partition, and finally drags the controller while pulling the trigger button. While dragging, a continuous partition surface is generated following the user's hand motion path (Figure 3 (b)). To delete the partition, the delete mode, indicated by the icon, shown in Figure 3 (c), can be selected, and the contact area between the icon and the partitions can be deleted.



Figure 3: User interface. (a) Menu, (b) Create mode: users can check the current partition status and create partitions (c) Delete mode: users can erase the part of partitions with the trash box pointer.

4 Use Scenarios

4.1 Workspace separation with free-form partitions

This system allows for the dynamic creation of various-sized partitions. Unlike physical partitions, which are limited by pre-fixed dimensions and physical constraints, AR partitions can be generated dynamically without such restrictions. Figure 4, shows two types of dominant partitions, desk partition, and room partition, respectively, which are interactively generated by our AR partition system. The distance of the partition (Figure 4 (a) Room) can be modified at the menu before generating.

To support more complex space needs, the transparency of the partitions can also be arbitrarily adjusted (Figure 4 (b)), which allows for maintaining spatial awareness while providing a clear physical boundary. More detailed user interface options are available in our video figure.

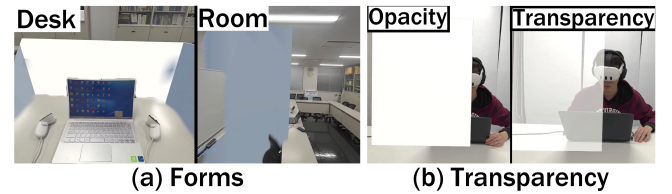


Figure 4: Examples of Custom AR partitions

4.2 Workspace modifications for ad-hoc interactions

One of the advantages of AR partitions is flexible modification capability. For example, as shown in Figure 5 (a), the user can instantly cut out the part of the located AR partition to make the user and others communicate in face-to-face formation. Once they have finished, the deleted partition can be restored to return to their original private spaces. Another example of supporting ad-hoc real-world communication is based on the proxemic interactions[1], where the content and partition functionalities change by the spatial relations among the users and content. For example, it is possible to make a brief face-to-face communication spot if one of the

users approaches another and penetrates the AR partition (Figure 5 (b)). They can be returned quickly by getting apart from each other. These are examples of how flexible AR partitions are while preserving real-world interaction capabilities.

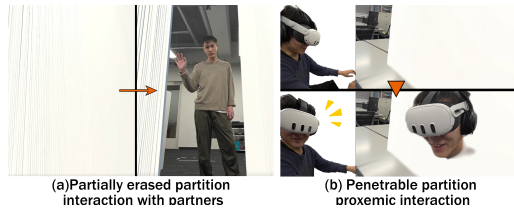


Figure 5: Example of ad-hoc real-world interactions

4.3 Multi-user AR meeting room

Our system supports multiple users sharing the same workspace. Figure 6 shows that one user is creating a virtual meeting room using a dome-shaped partition mode in a large, noisy co-working space, where a set of virtual screens are shared among multiple users in a visually and auditory quiet environment for the meeting.

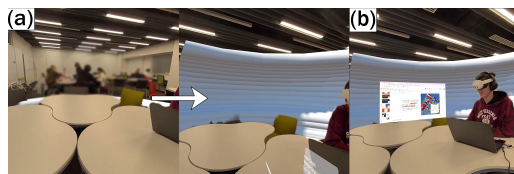


Figure 6: Private workspace for small-group work with AR Partitions

5 Limitations, Alternatives and Future work

We demonstrate a new AR partitioning system that allows multiple MR HMD users to introduce, edit, and delete the shared AR partitions arbitrarily according to their spatial needs and privacy demands. Since all users use the color pass-through MR HMD, they can manage both real and virtual elements simultaneously in quick and intuitive ways, giving new spatial design experiences and solutions for the current open-plan office problems.

However, current MR devices, such as the Meta Quest 3, still have limitations regarding resolution, especially when performing detailed tasks like reading fine print on real-world screens (e.g., documents on laptops). At present, virtual screens displayed by the HMDs offer a more comfortable viewing experience compared to physical screens. We expect next-generation AR/MR glasses or headsets could provide suitable alternatives to overcome this limitation.

Regarding the accuracy of ambient sound passthrough on noise-canceling headphones, while external sounds were easier to hear, there was a slight sense of discomfort compared to not wearing anything at all. Additionally, the precision of noise cancellation itself was not perfect to achieve complete sound isolation. In addition, the application we developed could not extract and acquire sound

sources. The reason was that a variety of sound sources exists in offices, such as voices, telephone ring tones, and machine operation noises. It is necessary to specify the sound source and discard these for better auditory noise canceling.

Future work includes conducting usability tests for our interface design and interpersonal psychological studies to evaluate how our AR partitioning system impacts communication management within open-plan office settings.

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References

- [1] Till Ballendat, Nicolai Marquardt, and Saul Greenberg. 2010. Proxemic interaction: designing for a proximity and orientation-aware environment. In *ACM International Conference on Interactive Tabletops and Surfaces (ITS '10)*. Association for Computing Machinery, New York, NY, USA, 121–130. doi:10.1145/1936652.1936676
- [2] Kazuyuki Fujita, Kazuki Takashima, Yuichi Itoh, and Yoshifumi Kitamura. 2023. Human-Workspace Interaction: prior research efforts and future challenges for supporting knowledge workers. *Quality and User Experience* 8 (2023), 1–19. Issue 1. doi:10.1007/s41233-023-00060-9
- [3] Rikiya Fukushima, Kazuyuki Fujita, Kodai Ito, and Yuichi Itoh. 2024. Visibility Control for Supporting Workers using Blind-based Partitioning Devices. *Quality and User Experience* 2024 9:1 9 (8 2024), 1–12. Issue 1. doi:10.1007/S41233-024-00067-W
- [4] M Itani, T Chen, T Yoshioka, and S Gollakota. 2023. Creating speech zones with self-distributing acoustic swarms. *Nature Communications* 14, 5684 (2023), 1–16. doi:10.1038/s41467-023-40869-8
- [5] Hyelip Lee and Andrea Bianchi. 2020. YourSphere: Supporting Flexible Environmental Switching in Open-plan Workspaces via Augmented Reality. *Archives of Design Research* 33 (2020), 43–52. Issue 4. doi:10.15187/ADR.2020.11.33.43
- [6] Hyelip Lee, Seungwoo Je, Rachel Kim, Himanshu Verma, Hamed Alavi, and Andrea Bianchi. 2019. Partitioning open-plan workspaces via augmented reality. *Personal and Ubiquitous Computing* 26, 3 (10 2019), 1–16. doi:10.1007/s00779-019-01306-0
- [7] Seung-Min Lee and Chan-Hoon Haan. 2023. Influence of Partition Layouts on Speech Privacy in Open-Plan Offices. *Heliyon* 10 (2023), 19. Issue 17. doi:10.2139/ssrn.4657346
- [8] Yuki Onishi, Kazuki Takashima, Shoi Higashiyama, Kazuyuki Fujita, and Yoshifumi Kitamura. 2022. WaddleWalls: Room-scale Interactive Partitioning System using a Swarm of Robotic Partitions. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. Association for Computing Machinery, New York, NY, USA, Article 29, 15 pages. doi:10.1145/3526113.3545615
- [9] Plutoberth. 2020. Sony Headphones Client. <https://github.com/Plutoberth/SonyHeadphonesClient>
- [10] Anastasia Ruvimova, Junhyeok Kim, Thomas Fritz, Mark Hancock, and David C. Shepherd. 2020. "Transport Me Away": Fostering Flow in Open Offices through Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3313831.3376724
- [11] Yuichiro Takeuchi. 2010. Weightless walls and the future office. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '10)*. Association for Computing Machinery, New York, NY, USA, 619–628. doi:10.1145/1753326.1753419
- [12] Sixuan Wu, Jiannan Li, Mauricio Sousa, and Tovi Grossman. 2023. Investigating Guardian Awareness Techniques to Promote Safety in Virtual Reality. In *Proceedings of the 2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR '23)*. 631–640. doi:10.1109/VR55154.2023.00078
- [13] Keng-Ta Yang, Chiu-Hsuan Wang, and Liwei Chan. 2018. ShareSpace: Facilitating Shared Use of the Physical Space by both VR Head-Mounted Display and External Users. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*. Association for Computing Machinery, New York, NY, USA, 499–509. doi:10.1145/3242587.3242630
- [14] Mary D Zalesny and Richard V. Farace. 1987. Traditional versus Open Offices: A Comparison of Sociotechnical, Social Relations, and Symbolic Meaning Perspectives. *The Academy of Management Journal* 30, 2 (1987), 240–259. <http://www.jstor.org/stable/256272>